

ILLUSTRATIVE EXAMPLES

✓ **Example 1 :** If $\sin \theta = -\frac{4}{5}$, $\pi < \theta < \frac{3\pi}{2}$, find
(i) $\sin 2\theta$, (ii) $\cos 2\theta$, (iii) $\tan 2\theta$, (iv) State the quadrant to which angle 2θ lies.

$$\sin \theta = -\frac{4}{5} \quad \pi < \theta < \frac{3\pi}{2}$$

Example 7 : If $\tan \frac{A}{2} = \frac{1}{\sqrt{3}}$, find $\sin A$.

$$\tan A = 8$$

Example 12 : Prove that : (a) $\frac{\sin \theta}{1 + \cos \theta} = \tan \frac{\theta}{2}$,

(b) $\frac{\sin 4\theta + \sin 2\theta}{1 + \cos 2\theta + \cos 4\theta} = \tan 2\theta.$ [S'12, 15]

$$= \text{R.H.S.}$$

Example 15 : Prove that

$$\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}.$$

$$\tan^2 2 = \text{R.H.S.}$$

Example 16 : Prove that : $\tan A \cdot \tan (60 - A) \cdot \tan (60 + A) = \tan 3A$.

[W'15]

Solution :

$$\text{L.H.S.} = \tan A \cdot \tan (60 - A) \cdot \tan (60 + A)$$

$$= 2 \cos \theta = \text{R.H.S.}$$

Example 20 : Prove that $\frac{\sec 4A - 1}{\sec 2A - 1} = \frac{\tan 4A}{\tan A}.$

Solution :

[W'13, 15]

$$\sec 4A - 1$$

$$(b) \frac{\cos 3\theta}{\cos \theta} + \frac{\sin 3\theta}{\sin \theta} = 4 \cos 2\theta$$

$$(c) \sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 8\theta}}} = 2 \cos \theta$$

$$(d) \frac{\sec 8\theta - 1}{\sec 4\theta - 1} = \frac{\tan 8\theta}{\tan 2\theta}$$

$$(e) \cos A \cdot \cos (60 - A) \cdot \cos (60 + A) \\ = \frac{1}{4} \cos 3A$$

$$(f) \frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} = \cot \frac{\theta}{2}$$

$$(g) \tan \theta + \sec \theta = \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$$

$$(h) \sqrt{\frac{1 - \sin 2\theta}{1 + \sin 2\theta}} = \tan \left(\frac{\pi}{4} - \theta \right)$$

Example 10 : Prove that :

$$\frac{\sin A + 2 \sin 2A + \sin 3A}{\cos A + 2 \cos 2A + \cos 3A} = \tan 2A. \quad [S'14]$$

Solution : L.H.S. = $\frac{\sin A + \sin 3A + 2 \sin 2A}{\cos A + 2 \cos 2A + \cos 3A}$

Example 12 : Prove that

[W'13, S'14]

$$\frac{\sin 7x + \sin x}{\cos 5x - \cos 3x} = \sin 2x - \cos 2x \cdot \cot x.$$

$$\sin 7x + \sin x$$

Example 13 : Prove that :

$$\frac{\sin \theta - \sin 3\theta}{\sin^2 \theta - \cos^2 \theta} = 2 \sin \theta.$$

[W'12]

Solution : L.H.S. = $\frac{\sin \theta - \sin 3\theta}{\sin^2 \theta - \cos^2 \theta}$

In this case, in numerator, use $\sin C - \sin D = 2 \cdot \cos \left(\frac{C+D}{2} \right) \cdot \sin \left(\frac{C-D}{2} \right)$ and in denominator, think $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$.

$$\begin{aligned} \therefore \text{L.H.S.} &= \frac{2 \cdot \cos \left(\frac{\theta + 3\theta}{2} \right) \cdot \sin \left(\frac{\theta - 3\theta}{2} \right)}{-(\cos^2 \theta - \sin^2 \theta)} \\ &= \frac{2 \cdot \cos 2\theta \cdot \sin (-\theta)}{-\cos 2\theta} \\ &= \frac{-2 \cdot \cos 2\theta \cdot \sin \theta}{-\cos 2\theta} \quad \dots \sin (-\theta) = -\sin \theta \\ &= 2 \cdot \sin \theta \quad \dots \text{Cancelling } \cos 2\theta \text{ and } \frac{-}{-} = + \\ &= \text{R.H.S.} \end{aligned}$$

Example 14 : Prove that :

$$\frac{\sin 4A + \sin 5A + \sin 6A}{\cos 4A + \cos 5A + \cos 6A} = \tan 5A.$$

[W'14, S'15, 16]

Solution : L.H.S. = $\frac{\sin 4A + \sin 5A + \sin 6A}{\cos 4A + \cos 5A + \cos 6A}$

Re-writing numerator and denominator for required R.H.S., we get

$$= \frac{(\sin 4A + \sin 6A) + \sin 5A}{(\cos 4A + \cos 6A) + \cos 5A}$$

Using appropriate factorization formulae, we have,

$$= \frac{2 \cdot \sin 5A \cdot \cos (-A) + \sin 5A}{2 \cdot \cos 5A \cdot \cos (-A) + \cos 5A}$$

$$= \frac{\sin 5A \{2 \cos A + 1\}}{\cos 5A \{2 \cos A + 1\}}$$

$\dots \because \cos (-A) = \cos A$

$$= \tan 5A = \text{R.H.S.} \quad \text{Proved.}$$

Example 15 : Prove that :

$$\frac{\cos 2A + 2 \cos 4A + \cos 6A}{\cos A + 2 \cos 3A + \cos 5A} = \cos A - \sin A \tan 3A.$$

[W'11, 13, S'12]

Solution :

$$\text{L.H.S.} = \frac{\cos 2A + 2 \cos 4A + \cos 6A}{\cos A + 2 \cos 3A + \cos 5A}$$

$$= \frac{\cos 2A + \cos 6A + 2 \cos 4A}{\cos A + \cos 5A + 2 \cos 3A}$$

... Observe carefully rewriting numerator and denominator

$$= \frac{2 \cdot \cos \left(\frac{2A + 6A}{2} \right) \cdot \cos \left(\frac{2A - 6A}{2} \right) + 2 \cos 4A}{2 \cdot \cos \left(\frac{A + 5A}{2} \right) \cdot \cos \left(\frac{A - 5A}{2} \right) + 2 \cos 3A}$$

... By factorization formula

$$= \frac{2 \cdot \cos 4A \cdot \cos (-2A) + 2 \cos 4A}{2 \cdot \cos 3A \cdot \cos (-2A) + 2 \cos 3A}$$

$$= \frac{2 \cos 4A \{ \cos 2A + 1 \}}{2 \cos 3A \{ \cos 2A + 1 \}} \quad \dots \cos (-\theta) = \cos \theta$$

$$= \frac{\cos 4A}{\cos 3A} \quad \dots \text{Cancelling } (\cos 2A + 1)$$

$$= \frac{\cos (3A + A)}{\cos 3A}$$

... Observe writing $4A = 3A + A$ carefully.

This is because of $\cos 3A$ in denominator

$$= \frac{\cos 3A \cdot \cos A - \sin 3A \cdot \sin A}{\cos 3A}$$

... By $\cos (A + B)$ expansion in numerator

$$= \frac{\cos 3A \cdot \cos A}{\cos 3A} - \frac{\sin 3A \cdot \sin A}{\cos 3A}$$

$$= \cos A - \sin A \cdot \tan 3A$$

$$= \text{R.H.S.}$$

Example 16 : Prove that :

[W'14]

$$\frac{\cos 3A + 2 \cos 5A + \cos 7A}{\cos A + 2 \cos 3A + \cos 5A} = \cos 2A - \sin 2A \tan 3A.$$

$$\text{Solution : L.H.S.} = \frac{\cos 3A + 2 \cos 5A + \cos 7A}{\cos A + 2 \cos 3A + \cos 5A}$$

$$= \frac{\cos 3A + \cos 7A + 2 \cos 5A}{\cos A + \cos 5A + 2 \cos 3A}$$

... Observe carefully rewriting

$= \text{R.H.S.}$

 **Example 21 : Without using calculator, prove that $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ = \frac{3}{16}$.**

[W'11, S'13, 14, 16]

Solution :

$$\begin{aligned} \text{(e)} \quad & \frac{\cos 3A - 2 \cdot \cos 5A + \cos 7A}{\cos A - 2 \cdot \cos 3A + \cos 5A} \\ &= \cos 2A - \sin 2A \cdot \tan 3A \end{aligned}$$